Self-Guided Evolution Strategies with Historical Estimated Gradients

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Motivation

In many optimization scenarios, the gradient information of objective functions can be unavailable.

Examples:

- Model optimization with discrete stochastic variables.
- Policy optimization in reinforcement learning.
- …

First Order Methods (SGD, Adam, …) **Zeroth Order Methods** (ES, …)

May fail !

- ✔ Perform well
- ✘ Inefficient with high variance in high-dimensional problems

We propose a new ES method SGES from the perspective of reducing variance, which has been verified to effectively solve high-dimensional optimization problems.

Evolution Strategies

Evolution Strategies(ES) are known as a class of black-box optimization algorithm, which usually consist of three steps:

Sample search directions from some distribution.
Estimate the gradient by a Monte Carlo estimator.
Update the parameters using the estimated gradient.

Vanilla ES [1]

Some distribution: isotropic multivariate Gaussian distribution

$$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_n)$$

Monte Carlo estimator: antithetic ES gradient estimator [2]

$$\hat{\nabla} f_\sigma^a(\boldsymbol{\theta}) = \frac{1}{2\sigma P} \sum_{i=1}^{P} \left(f(\boldsymbol{\theta} + \sigma \boldsymbol{\epsilon}_i) - f(\boldsymbol{\theta} - \sigma \boldsymbol{\epsilon}_i) \right) \boldsymbol{\epsilon}_i$$

SGES Algorithm

Gradient Subspace

Estimated gradients $\mathbf{G}$ $\xrightarrow{\text{SVD}}$ Gradient subspace $\mathcal{L}_{\mathbf{G}}$ → Corresponding to exploitation; Orthogonal complement $\mathcal{L}_{\mathbf{G}}^{\perp}$ → Corresponding to exploration

Main difference

$$\mathcal{P} = \begin{cases} \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_{\mathcal{L}_{\mathbf{G}}}) & \text{with probability } \alpha, \\ \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_{\mathcal{L}_{\mathbf{G}}^{\perp}}) & \text{with probability } 1-\alpha. \end{cases}$$

Adaptive Sampling

$$\alpha_t = \begin{cases} \min\{\delta \alpha_{t-1}, \kappa_1\} & \text{if } \hat{r}_{\mathbf{G}} \text{ does not exist or } \hat{r}_{\mathbf{G}} < \hat{r}_{\mathbf{G}}^{\perp}, \\ \max\{\alpha_{t-1}/\delta, \kappa_2\} & \text{if } \hat{r}_{\mathbf{G}}^{\perp} \text{ does not exist or } \hat{r}_{\mathbf{G}} \geq \hat{r}_{\mathbf{G}}^{\perp}, \end{cases}$$

where

$$\hat{r}_{\mathbf{G}} = \frac{1}{M} \sum_{i=1}^{M} \min\{f(\boldsymbol{\theta}_{t-1} + \sigma \boldsymbol{\epsilon}_i), f(\boldsymbol{\theta}_{t-1} - \sigma \boldsymbol{\epsilon}_i)\},$$

$$\hat{r}_{\mathbf{G}}^{\perp} = \frac{1}{P-M} \sum_{i=M+1}^{P} \min\{f(\boldsymbol{\theta}_{t-1} + \sigma \boldsymbol{\epsilon}_i), f(\boldsymbol{\theta}_{t-1} - \sigma \boldsymbol{\epsilon}_i)\}.$$

Measure the goodness of these two subspaces

Variance Reduction

Theorem 1. [3] Under certain assumptions, we have

$$\|\mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_n)}[\hat{g}_{ves}] - \nabla f(\boldsymbol{\theta})\| \leq \varepsilon,$$
$$|\mathrm{Var}[\hat{g}_{ves}] - (n+1)\|\nabla f(\boldsymbol{\theta})\|^2| \leq \varepsilon.$$

Theorem 2. Under certain assumptions, we have

$$\|\mathbb{E}_{\epsilon \sim \mathcal{P}}[\hat{g}_{sges}] - \nabla f(\theta)\| \leq \varepsilon,$$
$$|\mathrm{Var}[\hat{g}_{sges}] - \Omega| \leq \varepsilon.$$

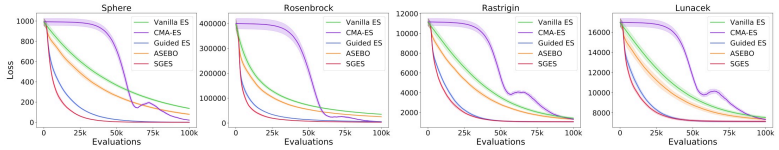
When the tradeoff α is well controlled and the representation of the gradient subspace is sufficient

$$\mathrm{Var}[\hat{g}_{sges}] \ll \mathrm{Var}[\hat{g}_{ves}]$$

The variance of the gradient estimator of SGES can be **much smaller** than that of Vanilla ES.

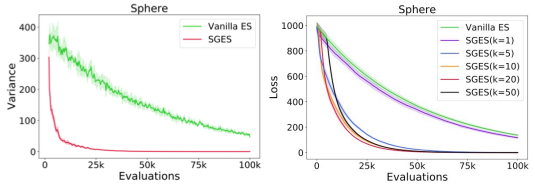
Experiment

Black-Box Functions Dimension = 1000



SGES achieves the best performance across all test functions.

Variance reduction Influence of k selection

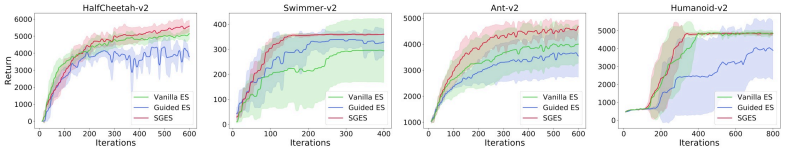


SGES can effectively reduce the variance.

SGES is not sensitive to k selection.

Experiment

Reinforcement Learning Tasks



Environment	Timesteps	TRPO	PPO	Vanilla ES	Guided ES	SGES
HalfCheetah-v2	10^7	2385	2876	4729	4032	5038
Swimmer-v2	$5 \cdot 10^5$	72	43	236	318	355
Ant-v2	$2 \cdot 10^7$	2037	2376	3260	2685	3927
Humanoid-v2	$3 \cdot 10^7$	2867	1325	3867	1765	4023

SGES achieves the best performance across all tasks.

[1] T. Salimans, et al. arXiv: 1703.03864, 2017.

[2] K. Choromanski, et al. ICML 2018.

[3] K. Choromanski, et al. NeurIPS 2019.